

TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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Abstract—Modern stereo estimators are applied independently to each frame and can fluctuate under specularities, weak texture, tissue deformation, tool occlusions and camera motion. We ask whether temporal information can improve frozen stereo predictions without modifying the estimator, accessing its internals, or using future frames.

We present TETHER, a causal plug-and-play temporal refiner. At each frame, preceding disparity is motion-aligned to the current view using frozen SEA-RAFT, and a shared 38-channel external evidence representation of disparity, motion and image geometry feeds a 154,874-parameter reset-and-fusion head predicting a bounded interpolation between the current estimate and that aligned memory. No learned appearance features and no frozen backbone enter that evidence; corrected recurrence is bounded and periodically re-anchored to independently predicted raw geometry.

A controlled development closure shows the learned policy is not explained by temporal smoothing: on SCARED-C D2, $H = 4$ reduces mean EPE by 3.69% against 0.59% for the best matched fixed/EMA blend and 0.87% for the best tuned unlearned residual rule, and those blends buy mean error with bad pixels while the learned gate does not. One frozen checkpoint improves all five evaluated estimators on held-out SCARED-C D7, including two excluded from temporal-head training, with mean backbone EPE $0.4921 \rightarrow 0.4645$ over three seeds ($5.61 \pm 0.42\%$; 0.4651 for the seed-0 checkpoint the tables report), and improves every one of 25 recording-backbone cells zero-shot on DRENDS. We additionally introduce a refinement-centric evaluation protocol measuring introduced error alongside recovered error under fixed, prediction-independent support, and apply it to our own claims.

I. INTRODUCTION

Metric three-dimensional perception underpins camera navigation, tool-tissue distance estimation, augmented reality and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth comes from images the platform already provides. It is also hard: endoscopic scenes carry weak texture, non-Lambertian tissue, specularities, blood, smoke, deformable anatomy, rapid viewpoint change and tool occlusion [18], [19]. Modern estimators [11]–[15] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods build temporal reasoning into the model [1]–[4], at the price of a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting: *can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?* Backbone and temporal logic can then evolve independently, a new estimator replacing the previous one without the module needing its cost volume, hidden state, confidence head or training

pipeline. We keep one inductive bias from the video-stereo literature — decide whether motion-aligned history should influence the current estimate, and constrain the correction to a convex interpolation between the two — and redesign everything around it for a black-box interface: the integrated motion pathway becomes frozen SEA-RAFT, backbone-specific representations become a shared external evidence space, and the same head is evaluated unchanged across architectures.

These restrictions raise a question: does the learned head add anything simple temporal averaging would not? Matched fixed fusion, EMA, warped-memory propagation and direct forward-backward confidence weighting recover only a small fraction of TETHER’s gain, while a ground-truth oracle shows the aligned memory carries far more than the policy exploits. The core problem is therefore not obtaining temporal evidence but deciding how much of it to use — and Sec. VI shows the head decides *how much* well and *where* no better than chance, which is what leaves the oracle gap open.

A second question is how such an intervention should be evaluated. Smoothness is not correctness, and lower mean EPE does not reveal whether refinement introduces new bad pixels or rare catastrophic frames, so we evaluate relative to the frozen upstream prediction and separate geometry, temporal fidelity, benefit, harm and tail behaviour.

Our contributions are three.

(i) We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a 154,874-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 38-channel external evidence space of disparity, motion and image geometry, support-confined convex fusion and bounded corrected-state recurrence.

(ii) We measure one frozen checkpoint across the full backbone $\times$ domain matrix — five estimators, two of them excluded from temporal-head training, over development, held-out and out-of-domain splits — including the joint unseen-backbone, unseen-domain cell that is usually left unmeasured, and it improves every one of the fifteen.

(iii) We introduce a refinement-centric protocol under fixed, prediction-independent support that scores introduced error alongside recovered error, and show it changes conclusions: under a matched state machine, every fixed or EMA blend that recovers a meaningful share of the gain buys it by *raising* the bad-pixel rate, a trade the learned gate does

not make and mean error does not reveal. Applied to our own claims it is equally unkind — degenerate history-replay controls win the no-reference consistency metric, and the head’s own weight inverts its relation to harm across splits.

II. RELATED WORK

a) Deep stereo matching.: Stereo has moved from cost-volume regression to iterative recurrent estimation [11], [12] and broad-domain zero-shot models [13]–[15]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD introduced the reset/fusion mechanism we build on [1]; DynamicStereo [2], PPMStereo [4] and TC-Stereo [6] follow, all co-designed with the estimator, so a new backbone means a new temporal model. Match Stereo Videos exposes BiDASStabilizer as a plug-in on a frozen estimator [3], but propagates bidirectionally. A parallel line enforces consistency on monocular video depth by diffusion [24] or stabiliser networks [16]: a different output space, none causal at clip level. A further line adapts the stereo weights online [25], the opposite of our constraint.

c) What can actually be compared.: We are aware of no published method that is simultaneously causal, black-box and estimator-agnostic. CODD [1], whose two-gate decision we build on, is causal but not black-box: its fusion network consumes matching features from its own stereo network, which a cached third-party backbone does not expose. The 142-channel configuration of Sec. V-I began as the closest reproduction of it our interface admits — its projection replaced by our warp, and a frozen ResNet-18 standing in for the features it would have taken from the estimator — so the comparison against it in Table IV is the nearest thing to running CODD here. Of the black-box candidates, StereoDiffusion [7] ships no implementation, and neural disparity refinement [8] and blind video temporal consistency [9] publish code whose weights are no longer retrievable; the supplement audits the set. Two methods are runnable here and we report both: BiDASStabilizer [3], which refines a frozen estimator as we do but bidirectionally, and TC-Stereo [6], an integrated architecture propagating its state by a rigid reprojection from per-frame pose — a privileged input our interface does not assume but SCARED-C supplies, so we run it as its authors intended (Sec. V-G).

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures. Recovered and introduced error must therefore both be measured, which connects to selective prediction [17] and to the uncertainty literature asking when a prediction should be trusted [26]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [18] and DRENDS [22] supply the endoscopic stereo and time-of-flight geometry we evaluate on; Sec. IV states what each can and cannot support. D4D [21] enters only as the substrate for a no-reference control, and SERV-CT [19] not at all: its pairs are non-consecutive, so temporal refinement on it is an exact identity and we report no result from it. Downstream reconstruction and tracking pipelines consume exactly the per-frame geometry we refine [20], [27], which is why we treat refinement as an intervention on that geometry.

III. METHOD

A. External Causal Interface

A frozen estimator S produces $d_t^{\text{raw}}, M_t^{\text{raw}} = S(I_t^L, I_t^R)$, positive-left disparity and its validity mask. The temporal module never reads the internal representation of S ; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is therefore $d_t^{\text{fused}} = T_\theta(d_t^{\text{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

B. Causal Motion Alignment

Frozen SEA-RAFT [5], [23] estimates both directions between the two already observed left images, $F_{t \rightarrow t-1} = F(I_t^L, I_{t-1}^L)$ and $F_{t-1 \rightarrow t} = F(I_{t-1}^L, I_t^L)$; the second is used only to measure reliability. Bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference.

Previous geometry is pulled onto the current grid, $\tilde{m}_t(x) = m_{t-1}(x + F_{t \rightarrow t-1}(x))$, by bilinear sampling with zero padding and `align_corners=True`. Warp support $\mathcal{S}_t$ records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way, $\tilde{F}_{t-1 \rightarrow t} = \mathcal{W}(F_{t-1 \rightarrow t}, F_{t \rightarrow t-1})$, gives the forward-backward residual $e_t^{FB} = \|F_{t \rightarrow t-1} + \tilde{F}_{t-1 \rightarrow t}\|_2$ and the reliability cue

$$C_t^{FB} = \mathcal{S}_t \exp(-e_t^{FB}/\rho_t), \quad (1)$$

$$\rho_t = 0.5 + 0.01 \left(\|F_{t \rightarrow t-1}\|_2 + \|\tilde{F}_{t-1 \rightarrow t}\|_2 \right).$$

The zero padding in this warp is not a detail: it is what makes $\tilde{m}_t$ exactly zero outside the source image, and Section V-J shows what happens when that value is allowed into the output.

C. Shared External Evidence

All estimators are mapped into one 38-channel tensor on the cache grid, built from raw disparity, aligned memory, the two images, the previous left image, the flow, C_t^{FB} , the warp support and the sampled historical validity. Twenty-five channels are disparity-geometry correlations formed at quarter resolution — 16 self-correlations of the two candidates and 9 raw-to-aligned cross-correlations over a dilated 3×3 neighbourhood — and 13 are formed on the grid directly: the two disparities, their signed and absolute

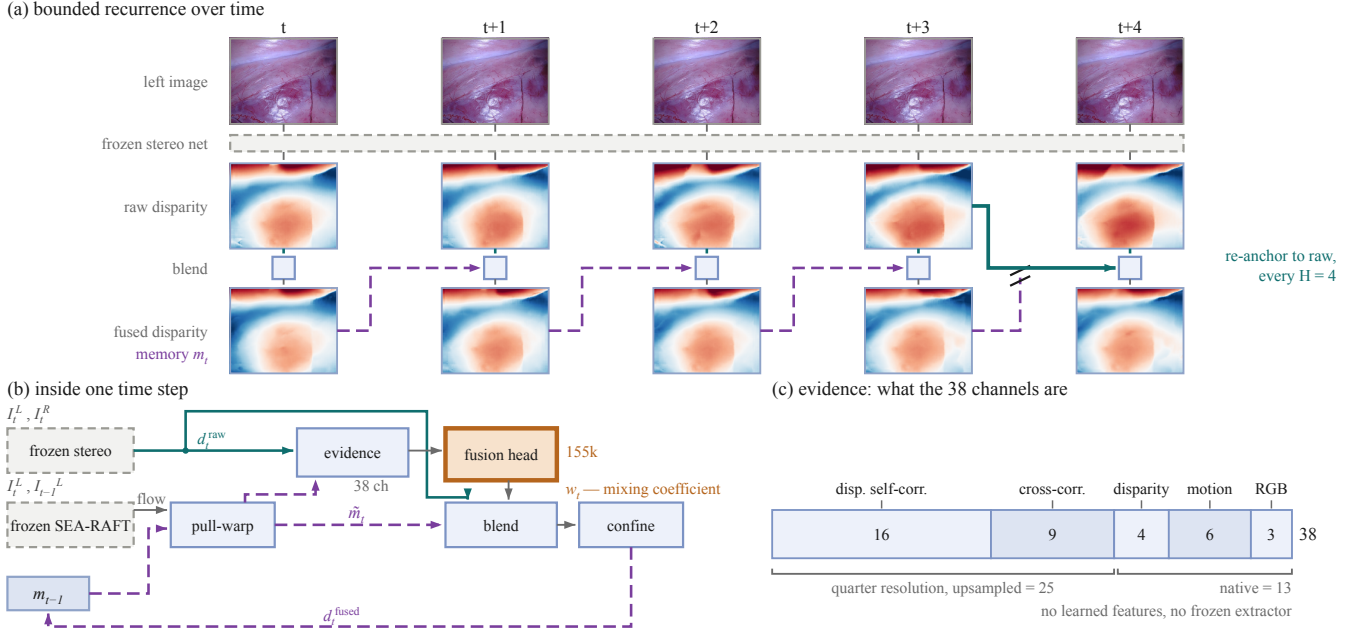


Fig. 1. TETHER at three scales. **(a)** Bounded recurrence: memory carries the *fused* disparity forward, and every $H=4$ frames the chain is cut and re-tied to raw geometry the frozen network produced independently (teal). The five frames are consecutive; the two disparity rows share one colour scale and differ by 0.126 px on average, with one fitted plane removed from all ten panels so the anatomy is visible under a 7 px depth ramp (the same plane from both rows, so their difference is untouched). **(b)** One time step; dashed grey is frozen, the single orange block is the only trained component and emits one scalar per pixel, w_t , rather than geometry. **(c)** The evidence space: none of the 38 channels carries ground truth, backbone identity or a future frame, and none comes from a learned extractor.

residual, 6 motion components and 3 RGB. Normalisation is fixed before training. Nothing in this space is learned, no network evaluates it, and no ground truth, backbone identity or future frame enters it. An earlier configuration added 104 quarter-resolution channels from a frozen ResNet-18 [10] for 142 in total; Sec. V-I reports it as an ablation, because it costs 157,504 frozen parameters and three forward passes per frame and does not survive a three-seed comparison. Exact channel construction and constants are in the supplement.

D. Reset-and-Fusion Decision Head

The two-gate decomposition of the temporal decision, and the constraint that the correction be convex in the two candidates, are CODD’s [1]; the architecture, the motion pathway, the evidence space, the supervision and the training are not.

A cache-grid pathway and a quarter-resolution context pathway feed a full-resolution reset branch predicting $r_t = \sigma(z_t^r)$ and a coarse fusion branch predicting $f_t = \uparrow_4 [\sigma(z_t^f)]$, giving the effective temporal weight $w_t = r_t f_t$ from 154,874 trainable parameters. Running the context branch coarsely is a compute choice costing $16\times$ less than full resolution, which Sec. V-I reports as an ablation. Layer widths are in the supplement.

Temporal evidence is only defined where the aligned memory is usable, so the per-pixel *support* is the conjunction of current raw validity, aligned temporal validity and warp support,

$$S_t = M_t^{\text{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \quad (2)$$

— the warp support $\mathcal{S}_t$ of the previous paragraph, the raw

validity and the sampled historical validity — and confine the intervention to it. Final disparity is

$$d_t^{\text{fused}} = \begin{cases} (1 - w_t)d_t^{\text{raw}} + w_t\widetilde{m}_t, & S_t = 1, \\ d_t^{\text{raw}}, & S_t = 0. \end{cases} \quad (3)$$

Because $w_t \in [0, 1]$ the fused disparity lies between the two candidates on the support, and off it the module is the exact identity on raw geometry: the refiner cannot synthesize disparity outside the interval they span. Applying (2) to the output, not only computing it, is load-bearing — the pull-warp’s zero padding makes an out-of-bounds sample $\widetilde{m}_t = 0$ exactly, and an unmasked blend there drives the output to zero (Sec. V-J). Reset output bias is initialized to -2 and fusion to zero, so initial intervention is conservative.

E. Bounded Corrected-State Recurrence

After refinement $m_t = d_t^{\text{fused}}$, so alignment and fusion errors can become future evidence. We bound corrected-state lifetime: the first output of a sequence is raw, a re-anchor restores previous raw disparity as state, and otherwise the previous fused output is propagated.

The shipped policy permits four corrected-state uses before a raw re-anchor. Thus H denotes state lifetime rather than a temporal input window.

The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \rightarrow d_t^{\text{fused}}$.

F. Training Objective

For target disparity g_t let $e_t^{\text{raw}} = |d_t^{\text{raw}} - g_t|$, $e_t^{\text{mem}} = |\widetilde{m}_t - g_t|$ and $\Delta_t = e_t^{\text{mem}} - e_t^{\text{raw}}$, so $\Delta_t < 0$ marks pixels where

aligned memory is the better candidate. Training support M is the intersection of GT coverage, current validity, aligned temporal validity and warp support.

The objective is $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$, unweighted, with $\mathcal{L}_{disp} = \mu_M[\text{Huber}_{\delta=1}(d_t^{\text{fused}} - g_t)]$, writing $\mu_A[\cdot]$ for the mean over pixel set A . The decision terms supervise the branches against Δ_t at native margins $\tau_r^n = 5$ px and $\tau_f^n = 1$ px, scaled to the cache grid by 0.140625: writing $R^\pm$, $F^\pm$ for the sets where Δ_t exceeds $\pm\tau_r^c$, $\pm\tau_f^c$ and F^0 for $|\Delta_t| \leq \tau_f^c$,

$$\mathcal{L}_{reset} = \mu_{R^+}[r_t] + \mu_{R^-}[1 - r_t], \quad (4)$$

$$\mathcal{L}_{fusion} = \mu_{F^+}[f_t] + \mu_{F^-}[1 - f_t] + 0.2 \mu_{F^0}[|f_t - 0.5|]. \quad (5)$$

No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. EXPERIMENTAL PROTOCOL

A. Training

The temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [18].

Training uses frozen predictions from S²M²-S [14], RAFT-Stereo [11] and Stereo Anywhere [13]; CREStereo [12] and Fast-FoundationStereo [15] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

Training uses D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs from three seen backbones, using non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0. The 154,874 trainable parameters are selected on development fused EPE. That budget deserves scrutiny, since selection fires at epoch 8, 11 and 11 across the three seeds and two of those are the penultimate epoch. What the curves show is a plateau rather than a climb: from epoch 5 the mean epoch-to-epoch movement in validation EPE (0.0025–0.0046 across the three seeds) is comparable to everything still gained after it (0.0041–0.0054), and all three seeds end the run above their own minimum. The two are not strictly comparable — one is a single step, the other a cumulative gain over several — so this supports the weaker claim that the epoch of the minimum is uninformative, not that the curve has converged. **[TODO: convergence probe: the shipped model retrained for 100 epochs against the locked 12, from training_runs/convergence_probe_A2_no_learned_evidence_100ep. Eight times the budget, because doubling it would only move the question. Not eligible for selection; if it keeps improving that is a limitation to report, not a model to adopt.]** One ablation (A4) diverges after epoch 10, which best-validation selection catches.

B. External Evaluation

a) *DRENDS*.: DRENDS contains dynamic robotic-endoscopy sequences with stereo imagery, calibration, and

metric geometry [22]. We report five curated recordings — Vid10_Liver_Med, Vid11_Liver_High, Vid12_Pancreas_Ext, Vid13_Pancreas_Med and Vid14_Pancreas_High — 7476 frames, evaluated with all five stereo estimators. The zero-shot claim applies only to the SCARED-trained checkpoint: no DRENDS data enters training, checkpoint selection or threshold calibration.

Disparity reference is derived from metric ToF geometry through stereo calibration. It is therefore useful metric evidence but not independent stereo ground truth, and the ToF reference itself is temporally filtered.

C. Unified Temporal-Refinement Evaluation

Evaluation support cannot depend on the prediction: $M_{\text{eval}} = M_{\text{GT}} \cap M_{\text{protocol}}$, and invalid predictions on valid support are penalised rather than silently removed. The framework scores five dimensions.

a) *Spatial geometry*.: EPE/MAE, median, RMSE, P90/P95/P99, maximum, invalid rate and Bad-1/3/5 on disparity; Bad-2/5/10 mm, AbsRel, SqRel and $\delta_{1,2,3}$ on metric depth.

b) *Temporal fidelity*.: At lags $k \in \{1, 2, 4, 8\}$, grid-based temporal change error DTCE = $|\Delta_{\text{prediction}} - \Delta_{\text{target}}|$ on fixed support, and, given correspondence, motion-compensated TEPE_{corr}, estimated-flow correspondences marked as proxies. DTCE scores agreement with the *reference* change, not stillness: a frozen prediction has $\Delta=0$ and is charged the whole of the reference motion, so the degenerate replay that wins the no-reference score of Sec. VII does not win this one. Every DTCE figure here is at $k=4$ unless stated. We do not build on the motion-compensated family: it improves on all five D7 backbones at all four lags but the two measures disagree on D2.

c) *Intervention harm and benefit*.: For $\delta e = e^{\text{fused}} - e^{\text{raw}}$ we measure the harm and benefit masses $H^+ = \mathbb{E}[\max(\delta e, 0)]$, $B^+ = \mathbb{E}[\max(-\delta e, 0)]$ and their rates HUR, BUR. Per bad-pixel threshold we report NewBad, RecoveredBad, bad-rate delta and identity preservation — the fraction of pixels already within τ still within it afterwards, a retention rate and not a count of untouched pixels, since the blend moves almost every pixel on the support. Per frame we retain the mean signed error change and its tails, and where a risk score exists, risk-coverage analysis [17]. Aggregation is named at every number: macro is the mean of per-cell relative reductions, pooled sums before the ratio.

V. RESULTS

A. Does the Learned Temporal Policy Matter?

We isolate the learned decision policy on D2. All variants share the frozen stereo predictions, the alignment, the support contract and the evaluation code; only the policy changes, and all run unconfined so the comparison shares one implementation (confinement, Sec. V-J, moves D2 EPE by 1.2×10^{-5} px). Figures here pool all five backbones and every D2 sequence before taking a ratio, so 0.276202 is one raw mean over the fifteen backbone-sequence cells; reductions

TABLE I

ONE FROZEN TETHER CHECKPOINT, SEED 0, ACROSS FIVE ESTIMATORS AND THREE ARENAS; EVERY EPE AND BAD1 CELL IS RAW→TETHER, LOWER IS BETTER. TWO ESTIMATORS WERE EXCLUDED FROM TEMPORAL-HEAD TRAINING, SO THE LOWER-RIGHT BLOCK IS THE JOINT SHIFT USUALLY LEFT UNMEASURED. DTCE IS THE MACRO REDUCTION IN TEMPORAL CHANGE ERROR AT LAG $k=4$, HIGHER IS BETTER; IT IS BACKBONE-DEPENDENT ON DEVELOPMENT AND REPORTED THERE IN SEC. V-B.

Backbone	Status	D2 development			D7 held out			DRENDS, out of domain		
		EPE	Bad1 (%)		EPE	Bad1 (%)	DTCE	EPE	Bad1 (%)	DTCE
S ² M ² -S	seen	.2704 → .2658	1.39 → 1.27		.5176 → .4992	6.44 → 6.03	6.5	1.3477 → 1.2812	62.55 → 58.98	11.3
RAFT-Stereo	seen	.2691 → .2623	1.60 → 1.43		.4671 → .4296	6.09 → 5.75	7.3	1.3125 → 1.2371	58.83 → 55.19	8.5
Stereo Anywhere	seen	.3022 → .2779	2.18 → 1.74		.4774 → .4408	6.23 → 5.77	7.9	1.3685 → 1.2834	62.73 → 57.99	13.9
CREStereo	unseen	.2668 → .2583	1.57 → 1.40		.5165 → .4912	6.35 → 6.05	6.4	1.3027 → 1.2296	55.17 → 52.21	11.8
Fast-FoundationStereo	unseen	.2725 → .2658	1.60 → 1.44		.4819 → .4645	5.92 → 5.65	4.3	1.2838 → 1.2228	54.88 → 52.06	9.6

TABLE II

MATCHED-POLICY CLOSURE ON D2, SEED 0, ALL FIVE BACKBONES AND ALL SEQUENCES POOLED BEFORE TAKING EACH RATIO; REDUCTIONS IN %, HIGHER IS BETTER, DTCE AT LAG $k=4$. EVERY ROW SHARES THE FROZEN PREDICTIONS, THE ALIGNMENT, THE SUPPORT CONTRACT, THE $H=4$ STATE MACHINE AND THE EVALUATION CODE — ONLY THE POLICY CHANGES, AND THE SIX ROWS ABOVE OURS LOAD NO CHECKPOINT. THE FULL GRID OF FIFTEEN POLICIES IS IN THE SUPPLEMENT.

Policy	EPE	Bad1	DTCE _{$k=4$}	NewBad1
Fixed $w=0.5$ ($\equiv$ EMA2)	0.59	-2.64	6.26	0.214
EMA, three frames	0.30	-8.88	4.88	0.379
FB-confidence as the weight	-2.17	-32.35	-7.10	0.831
Warped raw previous, $H=1$	-0.03	-6.39	-4.33	0.373
Aligned-memory propagation	-3.56	-41.70	-13.55	1.072
Unbounded corrected recurrence	-19.17	-143.48	-28.13	2.858
TETHER, $H=4$	3.69	12.84	4.99	0.063
Raw-vs-memory oracle (ceiling)	12.30	20.91	9.70	0.000

per backbone or per arena average per-cell relative reductions instead.

Raw D2 EPE is 0.276202. Table II gives the closure; the fifteen policies are the horizon grid $H \in \{1, 2, 4, 6, 8\}$ and continuous recurrence for the learned head, fixed $w \in \{0.1, 0.2, 0.3, 0.5\}$ rising monotonically to 0.59%, EMA over two and three frames, forward-backward confidence as the weight, aligned-memory propagation, and warped raw previous at $H=1$. A tuned two-parameter rule on the raw-versus-memory residual, swept over seventeen (α, τ) points, reaches 0.87% at an interior setting. Per backbone, against whichever fixed or EMA policy is best *for that backbone*, the ratio ranges $3.9\times$ to $9.9\times$ and is $5.0\times$ excluding Stereo Anywhere: the absolute gain depends on the estimator, the margin over smoothing does not.

Two rows carry more than the mean. Forward-backward confidence is the informative failure — one of TETHER’s own cues, yet used directly as the weight it worsens EPE by 2.17% and Bad1 by 32.35%. And the matched smoothing policies buy mean error with bad pixels, the more so the harder they blend: the two weakest fixed settings move neither metric by a hundredth of a point, $w=0.3$ costs 0.18% more bad pixels for 0.49% of EPE, the best fixed blend costs 2.64% for 0.59%, and EMA3 costs 8.88% for 0.30%. Averaging trades tail correctness for mean error as soon as it does real work; the learned gate does not make that trade,

and mean EPE alone does not show it.

B. Why $H = 4$, and Why Recurrence Must Be Bounded

Recurrence must be bounded: with no re-anchor the intervention inverts — EPE degrades by 19.17%, Bad1 by 143.48% and temporal stability by 28.13% (Table II). Within the bounded range EPE alone does not select an operating point — it improves from 2.49% at $H=1$ through 3.69% at $H=4$ to 3.73% at $H=6$ before falling to 3.52% at $H=8$. EPE does reject $H=1$, by 1.24 points; among the three horizons it does not separate, the spread is 0.21 — while newly broken pixels rise monotonically with the horizon, from 0.052% to 0.128%. We froze $H=4$ before D7 opened, on that risk-gain trade rather than on EPE, and it is the last horizon at which grid-based temporal change error also improves at lags $k \in \{2, 4, 8\}$: 4.99% at $H=4$ against -3.64% at $H=6$ for $k=4$.

Two caveats belong with that, because neither is visible in the pooled number. At $k=1$ DTCE improves at every horizon we tried, so the sign change is a property of the longer lags and not of the metric. And on D2 the improvement is carried by one backbone and is *negative* for three: +14.0% for Stereo Anywhere and +0.8% for CREStereo against -0.3%, -2.0% and -2.3% for the rest. On both confirmation splits it is positive for every backbone (Table I), which is why we report it there and rest the horizon choice on breakage rather than on stability.

C. Cross-Backbone Development Behaviour

$H = 4$ improves all five estimators in mean D2 EPE (Table I), from 1.41% to 8.19% averaging per-sequence reductions — 1.70% to 8.04% from the pooled cells the table prints — and all six unseen-backbone sequence groups improve; on D7 the mean excluding Stereo Anywhere is 5.15%. One D2 cell worsens, S²M²-S on keyframe 2, by 0.28%.

D. Held-Out Cross-Backbone Confirmation

We next evaluate the already frozen $H = 4$ policy on held-out D7. All five backbones improve, including both unseen during temporal-head training: mean EPE decreases 0.4921 → 0.4651, or 5.49% pooling before the ratio and 5.40% averaging per-cell reductions.

The improvement is not restricted to the mean: Bad1, RMSE and P99 decrease for every backbone on both splits,

so the EPE gain is not bought by a worse bad-pixel rate or a broader tail. New bad pixels do appear and are measured below. InvalidRate stays zero in all ten SCARED-C backbone-split evaluations, a property of SCARED-C and not of the module (Sec. V-J). The joint unseen-backbone, unseen-domain cell is the lower-right block of Table I.

E. Seed Variance, and What It Reveals About Selection

We pre-registered a seed study before any result existed: only the seed varies, sequences are the resampling unit, and selecting the best seed is prohibited.

Pooled over five backbones the relative Bad1 reduction is $12.73 \pm 0.32\%$ on D2 and $5.84 \pm 0.22\%$ on held-out D7; EPE gives $4.13 \pm 0.38\%$ and $5.61 \pm 0.42\%$, where $\pm$ is the standard deviation over the three seeds throughout this paper and *range* always means min-max. A paired bootstrap over the 15 and 20 backbone-sequence cells gives [8.96, 16.60] and [4.00, 8.71] for Bad1 and [2.55, 5.67] and [4.03, 7.28] for EPE, positive in every resample. We report those intervals as descriptive only: the cells share three and four underlying sequences, so they are not independent draws and the interval is narrower than the evidence warrants. The distribution-free statement the splits can carry is that the effect is positive in 3/3 and 4/4 sequences, a sign test at $p = 0.125$ and $p = 0.0625$.

The spread is the strongest argument for the shipped configuration, and it is not what we found first. The 142-channel ablation is stable on the held-out split (5.15 ± 0.34 Bad1) and unstable on development: its seeds fall monotonically over a 5.03-point range while the ordering inverts on D7 and the range collapses to 0.69. We first read that as selection optimism, since checkpoints are selected on best D2 fused EPE. Removing the learned evidence removes the instability instead, 0.32 points of D2 seed deviation against 2.53 with no change to the selection rule: the instability was the evidence space, not the protocol.

F. Refinement as an Intervention

Across both splits and all five backbones TETHER recovers $3.8\text{--}11.1\times$ as many Bad1 pixels as it introduces, B^+/H^+ between 1.39 and 3.25, identity preservation above 99.85%: it repairs far more than it breaks. Per frame, 56% of the 15,535 held-out frames improve against 14% worsened (45% against 18% on development), and the worsened band hugs the diagonal while the improved mass extends far from it — the count-versus-magnitude asymmetry behind B^+/H^+ . The scatter is in the supplement.

G. Head-to-Head Against a Published Stabiliser

BiDAStabilizer [3] is the closest published plug-in stabiliser we could run (Table III). We compare on DRENDS because only there is *neither* method in domain: we train on ex-vivo porcine tissue, it has never seen ex-vivo liver and pancreas manipulation against a time-of-flight reference. The shared input is its own RAFT-Stereo *robust* prediction, so TETHER is driven on that exact array over all five recordings (7476 frames) under one support neither method

influences. Two asymmetries run in opposite directions and both belong here. The baseline is bidirectional and consumes future frames, which our setting forbids. We, on the other hand, are trained on surgical endoscopy and it is running weights fitted to natural-image stereo, so “out of domain” is not symmetric: neither has seen DRENDS, but only one of us has seen this kind of scene. The same holds of TC-Stereo below. We therefore read the margin as an upper bound on our advantage, and an in-domain baseline remains the comparison we cannot run.

TABLE III

OUT-OF-DOMAIN COMMON SUPPORT, FIVE DRENDS RECORDINGS, 117.4M PIXELS; *neither* METHOD HAS SEEN THIS DOMAIN. EVERY ROW REFINES THE SAME FROZEN RAFT-STEREO *robust* PREDICTIONS; LOWER IS BETTER. $DTCE_k$ IS THE TEMPORAL CHANGE ERROR AT LAG k , IN PX, ON THE SAME SUPPORT. THE TWO BEST COLUMNS BELONG TO DIFFERENT ROWS, AND THAT IS THE FINDING: THE BIDIRECTIONAL BASELINE IS THE MORE TEMPORALLY STABLE METHOD AND THE LESS ACCURATE ONE.

Method	Causal	EPE	Bad1 (%)	Bad3 (%)	RMSE	DTCE ₁	DTCE ₄
Raw	—	1.199	44.92	4.54	1.876	0.242	0.340
BiDAStabilizer	no	1.175	44.81	4.32	1.776	0.085	0.214
TETHER	yes	1.131	43.39	3.94	1.710	0.139	0.294

The regime is hard and the Bad1 column says so: DRENDS ground-truth disparity averages 9.89 px, so a raw EPE of 1.199 is a third of the scene’s disparity spread and a 1 px threshold rejects 44.9% of pixels. What follows compares two refiners of a weak prediction, not a claim that either recovers a good one.

Out of domain for both, TETHER removes more error than the bidirectional baseline on all four metrics we score, while being the only causal entry and carrying $62\times$ fewer trained parameters (155k against 9.64M). That ratio is about training cost, not deployment: our system also runs frozen SEA-RAFT, 8.88M parameters and 26.2 of our 28.9 ms, so at system level the two are comparable and we make no efficiency claim. The exception is worth naming: on Vid13 TETHER is 0.022 px behind the baseline and 0.013 px *worse than the raw prediction it refines*, the one recording of five where the intervention is a net harm and the bidirectional method is not. In domain the ordering is the same and much wider — on D2, EPE $0.3029 \rightarrow 0.2889$ and Bad1 $2.09 \rightarrow 1.80\%$ against the baseline’s 0.3021 and 1.99% — but that margin is measured where we are in domain and it is not, which is why we report the neutral arena instead.

The temporal measure reverses the ordering, and we report it for that reason. Change error falls 64.9% for the bidirectional baseline against 42.5% for us at lag 1 and 37.2% against 13.8% at lag 4, in each of the five recordings and at every lag; in domain on D2 the split is the same, 52% against 15%. A method that sees the future is markedly more temporally stable than one that cannot, which is what should happen, and it is simultaneously the less accurate of the two on every geometric metric here. That is the paper’s own warning turned on its own comparison: a reader given

only a consistency score would rank these two the wrong way round.

TC-Stereo [6], the integrated-architecture reference, we run as its authors specify — released SceneFlow weights, its own iteration count, its rigid 6-DoF pose warp on SCARED-C’s corrected poses, with the pose convention validated against two wrong alternatives to 0.02px. On the same D2 support its frame path alone scores EPE 0.524 and Bad1 9.05% against 0.303/2.09% raw, so it is already outside its domain before any temporal component runs; enabling the temporal path makes both worse (0.622, 10.37%) and raises change error from 0.073 to 0.406px at lag 1, four to six times *less* consistent at every lag. We read that as a co-designed architecture meeting a scene its rigid warp does not describe, not as a weak method: a TC-Stereo trained on deforming tissue is an experiment nobody has run, ours included. Full protocol, self-check and per-sequence table in the supplement.

H. External Zero-Shot Geometry on DRENDS

We evaluate the frozen checkpoint on five DRENDS recordings with all five estimators, two excluded from temporal-head training, 7476 frames each. Nothing is adapted, retrained or tuned for this domain. Disparity EPE, Bad3, metric depth MAE and metric depth RMSE improve in *every* one of the 25 recording–backbone cells, by -5.40% , -19.51% , -4.02% and -3.47% on average. The two unseen estimators are not distinguishable from the three seen ones, which is the claim.

Residual variance is set more by the recording than by the estimator, but not uniformly: across the five backbones the EPE change spans 0.87 points on Vid10 and 1.03 on Vid11, against 3.18 on Vid12 and 3.29 on Vid13. Where the estimator matters, it matters as much as the recording does.

Averaged over backbones, the mean EPE reduction is 3.96% for seen and 2.67% for unseen estimators on D2, 5.80% and 4.79% on held-out D7, and 5.57% and 5.16% on DRENDS: every combination is positive. We do not read an ordering into it. On D2 the seen advantage is carried entirely by Stereo Anywhere, and without it the seen mean falls to 1.85% against 2.67% unseen — the ordering reverses. Five backbones cannot support a claim about which axis costs more, and we withdraw the one we drew. This establishes zero-shot geometric transfer under a ToF-derived metric reference; it does not establish universal cross-domain robustness, and DRENDS remains a non-independent reference.

I. Ablating Our Own Design Decisions

Four design choices were ablated under a pre-registration naming held-out Bad1 as the endpoint and the three-seed spread (0.37 points) as the threshold below which a difference is not read; A4 remains a single run. A separate later registration governs *promotion* of a variant to the shipped model and decides that on development only, which is how the 38-channel head was promoted; the two rules are distinct and we apply each where it belongs.

TABLE IV

HELD-OUT RELATIVE REDUCTION (%), NOT ERROR: HIGHER IS BETTER.

THE FOUR DEVIATIONS WERE PRE-REGISTERED AGAINST THE 142-CHANNEL CONFIGURATION; DROPPING ITS LEARNED EVIDENCE IS WHAT TETHER NOW IS, SO THE SHIPPED MODEL IS THE TOP ROW AND THE CONFIGURATION IT WAS ABLATED FROM IS THE ROW BELOW, WITH ITS DEVIATIONS INDENTED. THREE-SEED MEANS WHERE MARKED.

Configuration	Evidence	D7 EPE red.	D7 Bad1 red.
TETHER (3 seeds)	38 ch	5.61 ± 0.42	5.84 ± 0.22
+ learned evidence (3 seeds)	142 ch	5.06	5.15 ± 0.34
no appearance	78 ch	4.93	4.88
full-res context (3 seeds)	142 ch	5.24 ± 0.52	5.68 ± 0.16
unconstrained output	142 ch	0.05	6.02

The result (Table IV) is not the one we expected, and it is why the shipped model is the smaller one. We began where the video-stereo literature does, with a frozen ResNet-18 [10] supplying appearance correlations and cost curves, 104 channels on top of the 38 geometric ones — the configuration the four deviations were pre-registered against. Three of them sit above it on the held-out endpoint, so none of the decisions they undo is load-bearing. Removing the learned evidence entirely was then given the same three-seed treatment, under the promotion rule that decides on D2 with D7 reported afterwards, and it stays ahead on both endpoints with seed deviation eight times tighter (0.32 against 2.53 points of D2 Bad1). Out of domain the ordering holds: on the three backbones carrying three seeds of each, DRENDS EPE falls by $5.20 \pm 0.13\%$ against $4.83 \pm 0.15\%$, ranges disjoint at a permutation $p = 0.10$. The learned evidence is not what makes the module work; it is what makes its training unstable, at 157,504 frozen parameters, three forward passes per frame and 13.8% of the runtime. Its apparent advantage on the development closure (4.05% against 3.69%) is a seed-0 artefact: over three seeds TETHER leads there too, 4.13% against 3.55%.

Two rows need the registered rule read out loud, because it does not favour us. Running the context branch at full resolution reduces held-out Bad1 by $5.68 \pm 0.16\%$ against $5.15 \pm 0.34\%$ over three seeds each: it *wins* the registered endpoint by 0.54 points against a 0.37-point threshold, an outcome the registration did not enumerate. We decline it on cost rather than accuracy — $16\times$ the context-branch compute — which the registration required us to report if this happened. Ablating convexity (A4) posts the largest held-out Bad1 reduction in the table, 6.02%, on one seed; we reject it on EPE, a metric the registration names as reported but not as the endpoint, and that reading was chosen after seeing the result. Its registered consequence we do honour: the falsification clause required that if A4 matched or beat the canonical model held out, the claim attributing cross-backbone transfer to the convex action space be withdrawn rather than softened. It is withdrawn, and appears nowhere in this paper.

J. The Support Contract Is Load-Bearing

The results above use the support-confined output of (3); the ablation is a single line, with checkpoint, head, evidence, policy and reset schedule identical. It was measured on the 142-channel configuration, and concerns the output mask and the pull-warp padding rather than the evidence space. On DRENDS the difference is categorical. Unmasked, depth RMSE *regresses* in every cell evaluated, by +110.4%; confined, it improves in every cell, by −3.2%. The two arms are not scored on the same set — the unmasked run covers 14 of the 15 backbone-recording cells, missing RAFT-Stereo on Vid14 — so the comparison is directional and the magnitudes are not paired. Disparity follows more quietly, EPE from −4.30% to −5.00%, and invalid refined pixels go from up to 1.3×10^{-5} to exactly zero.

The cause is that zero-padded out-of-bounds memory enters the blend as $\tilde{m}_t = 0$, the fused disparity collapses there, and bounded recurrence carries it to the next re-anchor. The offenders are rare and bursty — 300 of 38.8M pixels on one recording, in runs of at most H frames — and the aligned memory of every one is exactly 0.0. Since an invalid prediction on valid support is penalised rather than discarded, those few hundred pixels are the entire source of the depth tail previously attributed to the external domain.

On SCARED-C the same ablation is nearly inert: D2 unchanged at 4.05%, D7 from 4.64% to 4.57%. The contract costs a little in-domain gain and buys well-posed behaviour out of it, and is almost invisible to anyone validating in domain only.

VI. ANALYSIS

Alignment is necessary but not sufficient: the aligned memory carries a 12.30% oracle ceiling that indiscriminate use does not collect (Table II). The mechanism is alignment plus learned graded intervention plus bounded recurrence, not smoothing, and the head never solves stereo matching — it receives two meaningful candidates and chooses between them.

a) *What each branch of $w_t = r_t f_t$ does.:* Each factor can be suppressed at inference with the trained head unchanged, which separates what the reset gate contributes from what the fusion weight contributes. Fusion alone keeps most of the gain, 3.49% against 3.69%, but breaks a third more pixels (0.084% NewBad1 against 0.063%) and drops B^+/H^+ from 1.85 to 1.72. Reset alone keeps 2.23% and breaks 0.211%, more than three times as many. The two factors therefore do different jobs: the fusion weight sets how much is gained, the reset gate sets how little is broken, and a paper reporting only the product and only mean error would have seen neither. On the 142-channel configuration the same decomposition is flatter in EPE (4.01% against 4.05%) and the count asymmetry is the whole of it, so the reset gate earns more of the mean gain in the shipped head than in the one it replaced.

Ablating the convexity constraint (A4) collapses mean EPE gain to 0.05% while improving RMSE and worst case: the failure is broad degradation of pixels already close, not

occasional large error, so the constraint buys mean accuracy rather than safety from large corrections.

Two limits are worth stating. Against the raw-versus-memory oracle of Table II, TETHER recovers 30.0% of the available gain on D2 as a ratio of the pooled quantities, 32.9% averaging per-cell recovery, so the open problem is deciding *where* to intervene. And the effective weight w_t , recovered from the convex form, is an intervention magnitude rather than an uncertainty: it ranks harmful intervention at AUROC 0.393–0.408 on D2 and 0.488–0.507 on held-out D7, below chance on the split it was tuned on and at chance on the split it was not. The 142-channel configuration is the same shape and further from chance (0.366–0.381 and 0.496–0.506), so this is a property of the formulation and not of the evidence space. It does not contradict the intervention ratios, which factorise into a count and a magnitude term needing no per-pixel ranking: on this analysis’ own support B^+/H^+ is 2.03 on D2 and 2.86 on D7, and per-cell oracle recovery rises from 32.9% on D2 to 42.7% on held-out D7. A policy can therefore be good on average and uninformative per pixel, which is what makes the oracle gap a ranking problem rather than a capacity one.

VII. DISCUSSION AND LIMITATIONS

a) *What the splits support.:* The closure is development work on D2: it supports mechanism analysis and the pre-frozen $H = 4$ operating point, not confirmation. Transfer is shown along both axes and their combination, but one external domain does not establish universal robustness, and DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth. Per-backbone D7 margins come from one seed and the three-seed study bounds only the pooled effect.

b) *What the metrics cannot say.:* No-reference temporal consistency is gameable, which we show on D4D [21] — deformable surgical stereo whose geometric scale contract is unresolved, so we make no geometric claim on it and use it only as the substrate for this control. The module reduces the no-reference score by 53.9%, but copy-previous removes 97.3% and warped-previous scores exactly zero by construction. Pull-warp support detects out-of-frame sampling but not internal disocclusion.

c) *What does not improve downstream.:* SCARED-C’s corrected poses let geometry be accumulated in one world frame and the disagreement between views of a point measured — what a map fuses rather than what a frame scores. On held-out D7 refinement reduces that disagreement in 4/4 sequences and for both heads, by 11.7% of the excess over a ground-truth floor of 0.06–0.21 mm (9.9% for the 142-channel ablation); on development D2 it *increases* it by 15.5% (7.1%). The splits differ in how much there is to remove — raw excess 0.55–1.11 mm on D7 against 0.18–0.22 on D2 — which suggests alignment recovers geometry a map can fuse where the raw prediction is poor and costs more than it gains where it is not. The reversal is not an artefact of one estimator: across five backbones the sign never changes, all D2 cells negative (−32.1 to −5.4%) and

all D7 cells positive (+8.7 to +15.4%). It does not settle the mechanism — the two raw-excess ranges stay disjoint, so no backbone makes D2 as hard as D7, and within D2 the ordering does not hold either. We report the condition as robust across estimators and do not claim raw difficulty as its cause. Mapping benefit is conditional, not general.

d) Cost, and where it is.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes: 28.90 ms overhead at 95.2 MB peak VRAM on an H100. The frozen estimator dominates that budget by up to an order of magnitude, so the end-to-end rate is set by the backbone, from 11.6 FPS on Fast-FoundationStereo (overhead 33.5%) to 1.4 FPS on StereoAnywhere (4.0%); *no real-time claim is made.*

Stage timing locates it: the two SEA-RAFT passes are 26.2 of the 28.9 ms, against 0.67 for the evidence tensor and 1.00 for the head. The evidence tensor is that cheap because TETHER builds no learned features — the 142-channel ablation spends 5.13 ms there, 13.8% of its total. The reverse pass exists only to compute C_t^{FB} , so dropping it would free 45% of the module; substituting a constant at inference moves the D2 closure from 3.69% to 3.74–3.79%, well inside the seed spread, so this head appears not to use the cue. We report it rather than ship it: it is development-only, on a head trained with the cue present, and a head trained without it is an experiment we have not run. Details in the supplement. We evaluate perception geometry, not downstream control, and demonstrate neither clinical safety nor patient benefit.

VIII. CONCLUSION

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: matched fixed and EMA blends under an identical state machine recover a sixth of it and buy that with bad pixels, a trade the learned gate does not make. One frozen $H = 4$ checkpoint improves all five estimators on held-out D7 and all 25 recording–backbone cells zero-shot on DRENDS.

Five results bound what may be claimed, and they are the reason the evaluation protocol is part of the contribution. No-reference consistency is won outright by replaying warped history. The fusion weight inverts its relation to harm between splits, so it is an intervention magnitude and not an uncertainty. Selection optimism on development is a property of the evidence space rather than of the protocol. A module that declares a support must apply it to what it emits. And world-frame reconstruction consistency improves held out while degrading on development, for every backbone we tried, so mapping benefit is conditional on something we can demonstrate but not yet name. What remains open is *where* to intervene: the oracle over the module’s own action space is far from reached.

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